

EBN111 SEMESTER TOETS/TEST 1 (3 MAR 2016)

ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	MEMO	Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans		English				

Q1		Q2		Q3		Tot:	/45
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VRAAG 1 – QUESTION 1 (12)

1.1	a) $i(t=10\text{ s}) = 1\text{ mA}$	b) $i(t=20\text{ s}) = 0\text{ mA}$ (Turning point)	
1.2	a) $Q = 30\text{ mC}$ 15 mA	b) $P = 16,2\text{ }\mu\text{W}$	
1.3	a) $I = 12,5\text{ A}$	b) $Q = 750\text{ C}$	c) $R = 360$
1.4	$P_1 = -18\text{ W}$	$P_2 = -16\text{ W}$	$P_3 = 0$
	$R_1 = 2\text{ }\Omega$	$R_2 = 4\text{ }\Omega$	

VRAAG 2 – QUESTION 2 (18)

2.1	a) $I_1 = 1,5\text{ A}$	b) $I_x = 4,7\text{ A}$
	c) $V_{5A} = 76\text{ A}$	d) $R = 20\text{ }\Omega$
2.2	a) $v_4 = -16\text{ V}$	b) $I = 4\text{ mA}$
2.3	a) $I_{40} = 2\text{ mA}$	b) $I_{10} = -8\text{ mA}$
2.4	$R_{ab} = 48\text{ }\Omega$	
2.5	$G_{ab} = 12\text{ mS}$	
2.6	$R_{eq} = 660\text{ }\Omega$	

VRAAG 3 – QUESTION 3 (16)

3.1	$I_X = 1,33\text{ A}$			
3.2	$I_Y = -0,5\text{ mA}$			
3.3	$V_Z = 8,67\text{ V}$ 7 V			
3.4	$i_0 = -0,6\text{ A}$			
3.5	$\alpha = 3$	$\beta = 5$		
3.6	$a = 5$	$b = 5$	$c = -0,5$	$d = -1$